

9-1-1 CALL PROCESSING & SYSTEM AUDIT

OPERATIONAL ASSESSMENT & SYSTEM COMPLIANCE TOOL

EXECUTIVE OVERVIEW

This audit workbook provides public safety administrators with a standardized template to evaluate emergency call handling and processing performance against national benchmarks. The metrics are grounded directly in the **NENA Standard for 9-1-1 Call Processing (NENA-STA-020.2-2026)** and the **NENA NG9-1-1 Call Processing Metrics Standard (NENA-STA-019.2-2022)**. Standardizing these measurements helps ensure consistency in emergency response across jurisdictional boundaries and protects agencies against service-related litigation by establishing clear, documented operational baselines.

INSTRUCTIONS FOR USE & METHODOLOGY

- 1. Monthly Data Evaluation:** The interval between Call Arrival and Call Answer must be evaluated monthly using a full month of continuous data to establish statistical validity. Performance metrics must be monitored and documented systematically.
- 2. High Impact Event (HIE) Exclusions:** High Impact Events refer to critical, unpredictable incidents that significantly affect public safety infrastructure or cause extreme surges in call volume (e.g., severe weather disasters, active violence). In accordance with NENA standards, HIE periods should be documented but **must be excluded** from standard performance compliance reporting to avoid skewing routine operational metrics.
- 3. Routine Transient Events:** Predictable, routine activity surges — such as daily commuter traffic, scheduled sporting events, concerts, and seasonal tourism — **must not** be excluded from performance reports. Staffing must be proactively managed to accommodate these predictable surges.
- 4. Complete Documentation:** Every attempt to provide public safety service must be fully documented in the CAD and telephony databases. This record-keeping serves as a critical shield against alleged failure-to-respond litigation.

Agency Name: _____

Assessment Period: _____

Lead Auditor: _____

Date Completed: _____

1. TELEPHONY & PHONE SYSTEM PERFORMANCE AUDIT

The telephony system captures signaling, session-state changes, and routing events. NENA standards define precise call-state changes to calculate critical connection and answering delays. The most critical benchmark is the **Call Answering Interval**, which measures the duration from **Call Arrival** (when the call is presented to the PSAP equipment) to **Call Answer** (when the call is delivered to and acknowledged by an agent or auto-greeting, and two-way communication begins).

Answering Priority & Protocols: On-duty telecommunicators must answer all incoming calls in strict order of priority: **(1) 9-1-1 or Alternate Emergency Access Numbers (AEAN)**, **(2) Non-emergency lines**, and **(3) Administrative/internal lines**. All 9-1-1 lines at a primary PSAP must be answered with the phrase '9-1-1' and agencies must verify the caller's exact location first. Location and callback verification are the absolute priority of initial call taking.

Abandoned, Disconnected, Silent & Non-Responsive Calls:

- *Abandoned Call:* Caller disconnects before answer. Telecommunicators must attempt to contact the caller once, at a minimum, in the same manner it was initiated (voice, text, or TTY) depending on capability.
- *Disconnected Call:* Disconnects prematurely after being received. Re-contact must be attempted at least once.
- *Silent & Non-Responsive Calls:* Must be immediately interrogated with a TTY/TDD device to ensure compliance with the Americans with Disabilities Act. Telecommunicators must monitor the line for background noises, and if an emergency is indicated, initiate public safety response immediately. A service disposition code must be added to the CAD record.

Performance Metric	i3 Log Event / Source	Technical Definition	Target Benchmark
Call Answer Interval	CallStartLogEvent to callAnswered State	Elapsed time between Call Arrival (presented to PSAP CPE) and Call Answer (agent or auto-greeting acknowledgment).	90% answered within <= 15 sec 95% answered within <= 20 sec
Abandoned Call Count	SIP CANCEL / callCancel LogEvent	Incoming emergency calls where the caller disconnects before an agent answers.	Document monthly; analyze inside & outside HIEs
Route Determination	CallStartLogEvent to RouteLogEvent	Time from call entering the Emergency Service Routing Proxy (ESRP) to routing determination.	Baseline established locally
Query Response Delay	AliLocationQuery / Response LogEvents	Round-trip delay between generating an ALI or location query and receiving verified location data.	Sub-second verification
Agent Availability	AgentStateChangeLogEvent	Time spent in primary states (Available/Not Available) before transition.	Performance tracking

2. COMPUTER-AIDED DISPATCH (CAD) SYSTEM AUDIT

The CAD system serves as the core database for logging public safety response activities, incident typology, and dispatch milestones. CAD metrics focus on operational efficiency once two-way communication has been established. The primary benchmark is the **Call Processing / Dispatching Delay**, which measures the interval from initial **Call Answer** to the moment the incident is **Dispatchable** (verified location, nature of emergency, severity, and responder safety information are collected and sufficient to notify responders).

NENA Call Processing Benchmarks:

- **High Acuity Emergency:** Emergency calls involving an imminent threat to life, serious injury, or significant risk to property (e.g., active violence, structure fires, hazmat) **should be processed and dispatchable within <= 60 seconds.**
- **Standard Emergency:** All other emergency calls **should be processed and dispatchable within <= 90 seconds.**

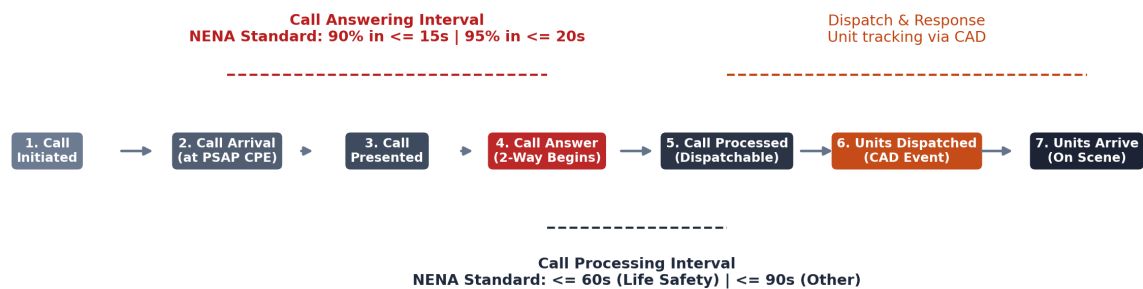


Figure 1.1: Complete NENA Emergency Call Process and Milestones

Performance Metric	CAD Milestone Code	Operational Definition	NENA / NFPA Benchmark
Call Processing (High Acuity)	Call Answer to Call Processed	Time to gather and verify location, chief complaint, severity, and life-safety risk for imminent threats.	90% processed in <= 60 sec
Call Processing (Standard)	Call Answer to Call Processed	Time to process and prioritize standard emergency incidents without immediate threat to life.	90% processed in <= 90 sec
Dispatch Delay	Call Processed to Units Dispatched	Interval between CAD incident entry/pre-alert and official notification/assignment of responder units.	Agency specific target
Response / Scene Arrival	Units Dispatched to Arrived Scene	Interval from unit notification/dispatch in CAD to physical arrival of the first unit on scene.	NFPA / Local policy standard
Service Disposition Code	Incident Disposition Entry	Adding resolution codes to CAD (accidental, welfare check, silent, resolved without dispatch) to track service quality.	100% database documentation

PSAP PERFORMANCE AUDIT WORKSHEET

Use this worksheet to calculate and document compliance for the assessment period. A full month of continuous database records is required. High Impact Events must be excluded from answering and processing statistics. Ensure all calculations are completed systematically.

Audited Operational Metric	Target Benchmark	Agency Performance (Period: _____)	Met?	Operational Corrective Actions / Notes
9-1-1 Call Answer Time (Arrival to Answer Interval)	>= 90% answered within <= 15 sec		☐ Y ☐ N	
9-1-1 Call Answer Time (Arrival to Answer Interval)	>= 95% answered within <= 20 sec		☐ Y ☐ N	
High Acuity Call Processing (Answer to Processed/Dispatchable)	>= 90% processed within <= 60 sec		☐ Y ☐ N	
Standard Call Processing (Answer to Processed/Dispatchable)	>= 90% processed within <= 90 sec		☐ Y ☐ N	
Abandoned Call Follow-Up (Attempt to re-contact)	100% of valid callback numbers attempted		☐ Y ☐ N	
Silent / Non-Responsive Calls (TTY / Text Interrogation)	100% TTY/Text-challenged under ADA		☐ Y ☐ N	
Location & Callback Verification (Standard Greeting Protocol)	100% active location verification first		☐ Y ☐ N	
Incident Service Disposition (CAD Documentation & Log)	100% database incident logging with codes		☐ Y ☐ N	

High Impact Event Exclusions Summary:

Total Excluded Events: _____ | Excluded Call Volume: _____ | Exclusions Reviewed & Approved: ☐ Yes ☐ No

Describe Excluded Incidents:

Operational Review Sign-Off:

I have reviewed the metrics and worksheet above and certify that the data is accurate and has been collected in accordance with NENA-STA-020.2-2026 methodology.

Auditor Signature: _____ Date Signed: _____